ATMA: Long-Context Language Modeling via Polar Attention and Gated-Delta Compression Memory

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Abstract

Long-context language models must balance retrieval, document likelihood, short-context quality, and inference cost. We introduce ATMA, a 378M-parameter hybrid that combines Polar Attention with gated-delta recurrent memory. Polar Attention separates a normalized direction from a bounded participation-ratio magnitude. In a complete 120-cell, 1B-token factorial sweep, memory improves Polar's 64K retrieval in all 20 matched cells by 47.8 points on average, while its effect on NoPE is small and inconsistent. We then train matched NoPE, RoPE, and Polar models for 9.816B tokens at length 2K and evaluate them through 256K. All models eventually lose exact retrieval on natural text, but Polar degrades later. It retains 62.7% exact accuracy at 8K and 18.0% exact synthetic retrieval at 256K. Polar also limits mean fixed-target bits-per-byte degradation to 1.26×, with a 1.9-point mean cost across eight short-context tasks. Separately optimized Raven baselines perform best on BABILong and use length-independent decode state. A post-hoc audit finds one near-unit retention gate in both NoPE and Polar, with zero-input half-lives of 21.0M and 3.07M tokens. Capping each outlying head at a 256-token half-life reduces mean 256K BPB from 8.097 to 1.595 for NoPE and from 1.825 to 1.501 for Polar. It also raises 256K BABILong from 0% to 39% and from 28% to 42%. We interpret excessive retention as a checkpoint-specific mediator, not a complete explanation or a validated training method.

1 Introduction

Large language models increasingly process code repositories, books, legal documents, and scientific papers that exceed their pretraining sequence length. Full softmax attention preserves token-level access in principle, but its probability mass can disperse as the key set grows (Nakanishi, 2025; Velickovic et al., 2025; Barbero et al., 2024). Sliding windows bound cost by removing older tokens from direct access. Recurrent sequence models instead keep fixed-size state but compress history. Long-context modeling therefore exposes a multi-objective trade-off among retrieval fidelity, sequence likelihood, reasoning, short-context quality, and serving cost.

On exact associative recall in natural text, every tested model eventually falls to zero under unadapted length extrapolation. Raven is near zero across the tested range, while RoPE and NoPE fall sharply by 4K and 8K. Polar Attention retains exact natural-text recall through 8K and some recall at 16K, with partial target-token signal through 256K.

We present ATMA, a hybrid architecture that combines short gated convolutions with four Polar Attention layers as global mixers. Polar Attention represents "what matched" with a normalized direction and "how much matched" with a bounded participation-ratio channel. A learned null sink follows the extreme-value scale of irrelevant scores. Each global layer also contains gated-delta memory, providing a compressed recurrent history alongside full-context attention (Appendix A).

We make three contributions. First, a complete 120-cell, 1B-token factorial sweep measures interactions among the attention core, recurrent memory, local training windows, distrac-

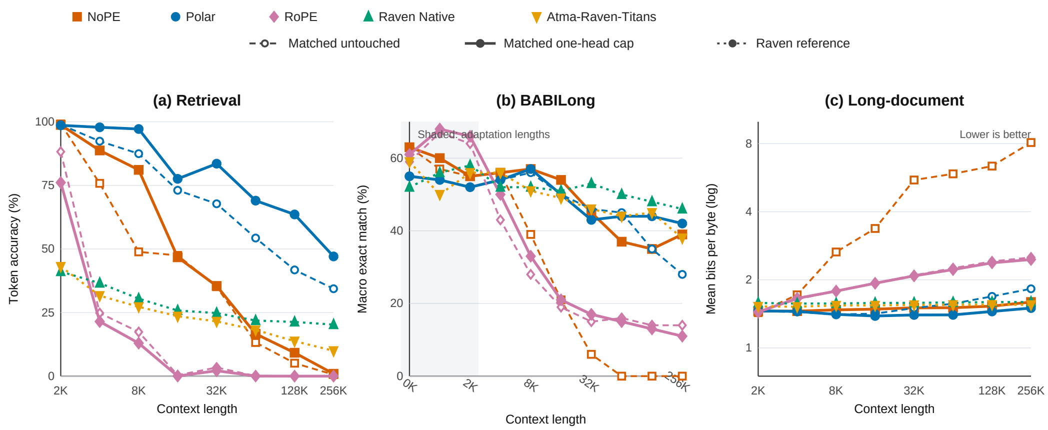

Figure 1: Results by context length. Dashed open curves show the original matched checkpoints, and solid curves show the inference-only one-head cap. Dotted filled curves show the untouched Raven references, which are neither capped nor optimizer-matched. Left: teacher-forced target-token retrieval. Center: BABILong exact match after adaptation. Right: mean fixed-target BPB on a log scale (lower is better).

tor alignment, and representation regularization. Second, matched 9.816B-token NoPE, RoPE, and Polar runs compare retrieval, BABILong, fixed-target likelihood, short-context tasks, and serving. Separately optimized Raven models provide recurrent-family reference points (Afzal et al., 2026). Third, we trace checkpoint-specific extrapolation failure from nearly identical 2K loss curves to isolated near-unit retention gates. A single-head, inference-only intervention tests this mechanism across all evaluated lengths. Its selective recovery supports excessive retention as a causal mediator in NoPE and Polar, while negative controls and remaining failures bound the claim.

2 Related Work

Softmax attention (Vaswani et al., 2017) has quadratic compute and can lose selectivity as the number of keys grows. RoPE (Su et al., 2021) and ALiBi (Press et al., 2021) alter positional information while retaining dense simplex normalization. Scalable-Softmax, adaptive-temperature analyses, and sparse alternatives address dispersion or length-dependent sharpness (Nakanishi, 2025; Velickovic et al., 2025; Vasylenko et al., 2026). Threshold Differential Attention (TDA) uses a $\sqrt{\log n}$ threshold motivated by maxima of noise scores (Huang et al., 2026). Polar combines length-dependent temperature and extreme-value scaling with a differentiable null sink and a bounded magnitude channel. We compare Polar with matched NoPE and RoPE controls. Comparisons with the other methods are conceptual rather than empirical.

Linear attention, state-space models, and gated recurrences exchange full-resolution token access for compressed fixed-size history (Gu & Dao, 2023; Yang et al., 2024b;a; Behrouz et al., 2025). Near-unit gates create very long timescales and can enter saturation regimes (Gu et al., 2020). Stability-aware SSM parameterizations likewise distinguish stable eigenvalues from a robust distance to the stability boundary (Wang & Li, 2024). Stuffed Mamba directly studies an inability-to-forget failure and post-training interventions that restore forgetting (Chen et al., 2025). Our audit asks the corresponding checkpoint-level question for the gated-delta branch used here: a sigmoid keeps each retention value below one, but does not enforce a useful uniform margin from one. Raven combines routed sparse memory, decay dynamics, and gated-delta recurrence (Afzal et al., 2026). We treat it as a strong sub-quadratic baseline, but not as an attention ablation because it uses a different model family and optimizer. Hybrid models such as LFM2 combine local and global mixers for efficiency (Hasani et al., 2025; Allen-Zhu, 2025). ATMA follows this hybrid pattern while changing the global mixer and adding a recurrent memory stream.

3 Method

3.1 Hybrid architecture

ATMA is a 16-layer decoder whose block order repeats local–local–global–local four times, yielding 12 LFM2-style gated convolutional layers and global attention at blocks 3, 7, 11, and 15. All blocks use pre-normalization and squared-ReLU MLPs. The attention variants share hidden size 1024, eight query heads, two key-value heads, head dimension 128, tokenizer, data order, optimizer, schedule, and training budget. Only the attention core changes among NoPE, RoPE, and Polar. Canon depthwise causal convolutions are applied to q, k, and v before attention (Allen-Zhu, 2025).

3.2 Polar attention

For query i and its $n_i = i + 1$ visible keys, let $\sigma_{ij} = q_i^\top k_j/\sqrt{d_k}$. Each head learns a length temperature and a virtual null logit,

$$t_i = 1 + \mathrm{softplus}(\alpha)\log n_i, \qquad \nu_i = b + \mathrm{softplus}(\gamma)\sqrt{\log(n_i+1)}. \tag{1}$$

The second term tracks the extreme-value scale of unrelated scores. Appending the null as a softmax competitor gives

$$w_i = \mathrm{softmax}(t_i[\sigma_{i\bullet}, \nu_i])\,, \qquad s_i = \sum_j w_{ij} v_j + w_{iN} v_{\mathrm{null}}. \tag{2}$$

The direction channel is $c_i = s_i/\max(\|s_i\|_2, \varepsilon)$, hence $\|c_i\|_2 \le 1$. To retain bounded information about match multiplicity, real-key weights are renormalized as $\hat{w}_{ij} = w_{ij}/\sum_k w_{ik}$ and summarized by the participation ratio

$$n_{\mathrm{eff},i} = \left(\sum_j \hat{w}_{ij}^2\right)^{-1}, \quad m_{\mathrm{eff},i} = n_{\mathrm{eff},i}(1 - w_{iN}), \quad \mu_i = \tanh(\mathrm{softplus}(\beta)\log(1 + m_{\mathrm{eff},i}))\,. \tag{3}$$

Thus $\mu_i \in [0, 1)$, while the direction channel removes radial scale. This is a channel decomposition, not an independence claim, because changing the match set can rotate c_i. The Polar output is a gated projection of c_i plus a learned projection of μ_i. Pre-integration tests rejected simpler choices. Additive soft counts leaked with length, fixed or relative floors were overtaken by noise tails, and an unbounded $\log m$ channel left the training range. The standalone prototype was not retained. Appendix C separates this design history from the reproducible integrated oracle and kernel tests. The promoted recipe disables the distractor-alignment loss tested in the factorial.

3.3 Gated-delta memory

Each global layer also maintains a per-head recurrent state $M_t \in \mathbb{R}^{d_v \times d_k}$. With unit-normalized keys and queries, retention $\gamma_t = \sigma(W_\gamma x_t + b_\gamma)$, and write gate β_t, the decay-first, self-inclusive recurrence is

$$M_t = \gamma_t M_{t-1}(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top\,, \qquad r_t = M_t q_t. \tag{4}$$

RMS-normalized r_t is gated, projected, and added to the Polar output. The projection is zero-initialized. Unit L_2 normalization is important. RMS-normalized keys have $\|k\|_2^2 = d_k$ and made the tested delta recurrence diverge, while unit keys keep the update eigenvalue along k within the gate-controlled range. Under bounded values and a uniformly contractive retention gate, Appendix A gives a sequence-length-independent state bound. The sigmoid guarantees $0 < \gamma_t < 1$ pointwise, but not the fixed margin $\gamma_t \le \bar{\gamma} < 1$ assumed by that result. The implied half-life $H_{1/2} = \log(1/2)/\log\gamma$ diverges as $\gamma \to 1$. Section 6 audits this distinction empirically.

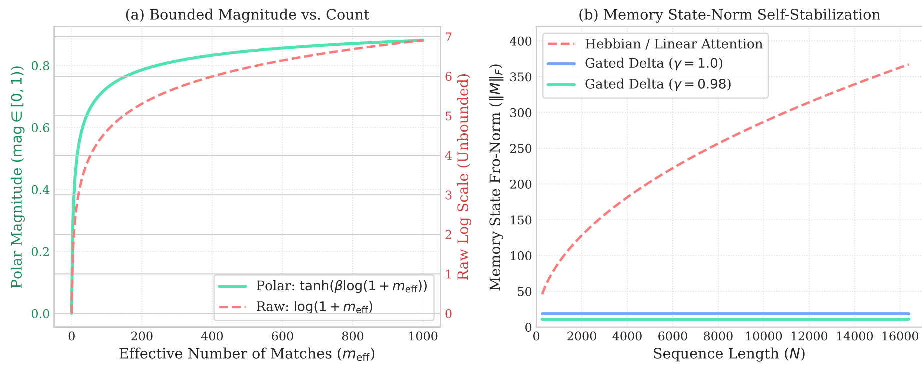

Figure 2: Controlled ATMA diagnostics. Left: the effective match count maps to a bounded magnitude channel. Right: the tested gated-delta state stays flat with length, unlike an ungated Hebbian memory. These diagnostics illustrate the mechanism but do not prove network-level stability.

Figure 2 summarizes the two mechanisms and their controlled diagnostics. Appendix B defines every regularization mode in the factorial sweep, Appendix C gives the numerical oracle and streaming Triton implementation, and Appendix H details the kernel and serving optimizations.

4 Experimental Design

4.1 Recipe selection

Stage I trains all $3 \times 5 \times 2 \times 2 \times 2 = 120$ combinations of attention type (Polar, RoPE, NoPE), regularization, distractor loss, memory, and a 1024-token training window. Every model receives about 1B FineWeb-Edu tokens at length 2048 on an NVIDIA L4, and every evaluation uses full context. These are distinct factorial cells, not seed replicates. Stage I selects the promoted recipe and is not used for headline absolute comparisons.

4.2 Matched comparison

Stage II trains 378.16–378.22M-parameter NoPE, RoPE, and Polar models for 9.816B tokens at length 2048 in the same L40S software and hardware environment. All three use the same Muon/AdamW parameter split. Raven Native (388.54M) and Atma-Raven-Titans (382.37M) use the same data, tokenizer, token budget, and device class, but use AdamW. In a 1B-token Raven Native pilot with the ATMA Muon schedule, validation loss improved to 3.76 nats at step 625, then became unstable and ended at 4.62. Retrieval remained at 0% through 64K. We therefore report the stable Raven models as a separate model-family/optimizer group. The single failed pilot does not establish incompatibility with every Muon configuration.

Evaluation. The evaluation contains 24,000 paired retrieval trials over synthetic and FinePDFs haystacks, lengths 2K–256K, depths 10/50/90%, and 50 trials per cell. It also includes BABILong adaptation and exact match (Kuratov et al., 2024), fixed-target bits per byte (BPB) on FinePDFs, PG-19, and Proof-Pile, eight likelihood-scored downstream tasks (22,939 examples per model), and single-sequence prefill and decode on one L40S. Retrieval teacher-forces the five-token target and reports both token accuracy and exact five-token accuracy. Neither metric measures autoregressive generation success. BABILong uses custom splits and prompts, so its absolute scores are internally comparable rather than leaderboard replications. Protocols, denominators, and complete matrices appear in Appendix D.

64K matched contrast	Mean Δ	Median Δ	Range	Positive
Polar: memory on − off	+47.8	+51.3	[+7.5, +92.5]	20/20
NoPE: memory on − off	+4.6	+1.9	[−15.0, +21.3]	11/20
RoPE: memory on − off	0.0	0.0	[0.0, 0.0]	0/20
Polar+memory − NoPE+memory	+41.7	+41.9	[0.0, +77.5]	19/20 (+1 tie)
Polar+memory − RoPE+memory	+50.3	+51.9	[+18.8, +92.5]	20/20

Table 1: Matched contrasts across the 120-cell, 1B-token factorial. Values are percentage-point changes in teacher-forced five-token accuracy at 64K. Each row holds all other factorial axes fixed. "Positive" excludes one numerical tie in the Polar–NoPE comparison.

Group	Model	Retrieval (%)			BABILong (%)		BPB		
		Tok. 2K	Tok. 256K	Exact 256K	2K	256K	2K	256K	ratio
Matched	NoPE	99.2 → 98.8	0.6 → 0.9	0.0 → 0.0	55 → 55	0 → 39	1.432 → 1.454	8.097 → 1.595	5.65× → 1.10×
Matched	Polar	98.9 → 98.6	34.4 → 47.1	9.0 → 16.3	52 → 52	28 → 42	1.451 → 1.458	1.825 → 1.501	1.26× → 1.03×
Matched	RoPE	88.2 → 76.1	0.0 → 0.0	0.0 → 0.0	64 → 66	14 → 11	1.439 → 1.440	2.512 → 2.460	1.75× → 1.71×
Raven/AdamW	Atma-Raven-Titans	43.1	10.0	0.0	56	38	1.525	1.551	1.02×
Raven/AdamW	Raven Native	41.1	20.3	0.0	58	46	1.581	1.597	1.01×

Table 2: Long-context endpoints. Matched attention variants show untouched → capped results, while Raven models are untouched reference points. Retrieval averages tasks, suites, and depths. "Tok." is teacher-forced target-token accuracy, and "Exact" requires all five target tokens to be correct. BABILong is macro exact match. BPB averages three fixed-target datasets (lower is better). Bold marks the best result within each comparison group.

5 Results

5.1 Component selection

Table 1 summarizes matched contrasts from the full factorial rather than selected cells. At 64K, adding memory improves Polar in all 20 combinations of regularizer, distractor setting, and window setting, by 47.8 points on average. Its effect on NoPE is small and mixed, and RoPE remains at zero. Among memory-enabled cells, Polar is never worse than NoPE and exceeds RoPE in all 20 comparisons. A training window reduces Polar+memory by 25.5 points on average and hurts 8 of 10 matched cells. Distractor alignment has a smaller, mixed effect, averaging −6.0 points and hurting 6 of 10 cells. We therefore promote full-context Polar plus memory without distractor alignment. The selected baseline cell reaches 92.5% token accuracy at 64K. Appendix E gives the complete matrix.

5.2 A multi-objective frontier

Table 2 reports the extrapolation endpoints. At 2K, NoPE and Polar have nearly perfect target-token accuracy. At 256K, NoPE and RoPE fall to 0.6% and 0.0%, while Polar retains 34.4% averaged across tasks, suites, and depths. Exact five-token accuracy gives a stricter picture. Raven Native scores 0.0% on FinePDFs from 2K to 256K, and Atma-Raven-Titans falls from 4.7% at 2K to 0.0% at 4K. RoPE falls from 77.3% at 2K to 0.0% at 4K, and NoPE falls from 96.7% at 2K to 0.0% at 8K. Untouched Polar retains 93.0%, 67.3%, 62.7%, and 12.0% on FinePDFs at 2K, 4K, 8K, and 16K. On synthetic haystacks, it retains 65.7% exact retrieval at 64K and 18.0% at 256K. With the retention cap, Polar stays above 91% exact FinePDFs retrieval through 8K, remains nonzero through 128K, and reaches 32.7% exact synthetic retrieval at 256K. Polar therefore delays, but does not eliminate, the loss of exact natural-text retrieval under length extrapolation. Appendix F gives the full token, exact, depth, and haystack matrices.

Table 3 separates target-token accuracy by haystack and length. Appendix F reports the corresponding exact-sequence values. Polar limits mean BPB degradation to 1.26× at 256K, compared with 5.65× for NoPE and 1.75× for RoPE. This stability comes with a short-context cost. Mean accuracy over eight 2K controls is 43.29% for Polar, 45.15% for NoPE,

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.1 → 98.5	98.4 → 97.5	97.1 → 96.1	97.0 → 96.0	94.9 → 93.9	92.3 → 92.5	80.3 → 85.5	68.4 → 77.5
	FinePDFs	98.6 → 98.7	86.3 → 98.1	77.9 → 98.3	49.1 → 59.1	40.7 → 73.1	16.3 → 45.5	3.2 → 41.6	0.3 → 16.7
NoPE	Synthetic	99.1 → 98.7	98.7 → 97.7	97.7 → 95.8	94.9 → 90.8	70.3 → 69.7	26.5 → 33.2	10.2 → 18.4	1.2 → 1.9
	FinePDFs	99.3 → 98.8	52.9 → 79.8	0.0 → 66.4	0.0 → 2.7	0.0 → 1.3	0.0 → 0.1	0.0 → 0.0	0.0 → 0.0
RoPE	Synthetic	81.7 → 63.4	44.9 → 40.3	34.5 → 25.9	0.8 → 0.1	6.8 → 4.3	0.1 → 0.1	0.0 → 0.0	0.0 → 0.0
	FinePDFs	94.8 → 88.8	4.7 → 2.7	0.2 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0
Atma-Raven-Titans	Synthetic	63.0	60.7	53.8	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	60.1	57.7	52.7	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 3: Zero-shot retrieval token accuracy (%) by haystack type. Matched models show untouched → capped results, while Raven variants are untouched references. Values average teacher-forced token accuracy over needle depths.

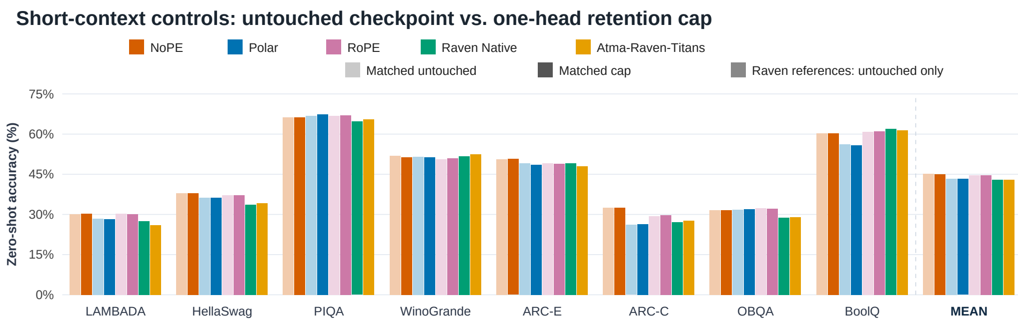

Figure 3: Zero-shot accuracy on eight 2K language-modeling controls (22,939 examples per model). For matched attention variants, pale bars are untouched and saturated bars are capped. The Raven models appear only as saturated, untouched references. The rightmost group gives unweighted task means. Table 20 reports the paired values for matched models.

and 44.60% for RoPE. Figure 3 shows that the gap varies by task. The 1.86-point mean gap to NoPE rules out a blanket quality claim. Appendix G gives the per-task values.

Table 4 shows the quality–systems trade-off. Raven Native reaches 46% BABILong at 256K and keeps BPB nearly flat. Its fixed recurrent state also avoids a sequence-length-dependent KV cache. At 128K, Atma-Raven-Titans decodes at 2.27 ms/token, compared with 16.3 ms/token for Polar. Polar instead has higher synthetic retrieval and training MFU. The BABILong samples are small. Simple Wilson intervals at 256K are 20.1–37.5% for Polar (28/100) and 36.6–55.7% for Raven Native (46/100). Because Raven also changes model family and optimizer, these rows show practical operating points rather than isolate one causal mechanism.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 4: Short-context and systems trade-offs. Serving uses one sequence and one timing sample on one L40S, so the values are descriptive rather than uncertainty estimates.

6 Checkpoint Variability and Retention-Horizon Audit

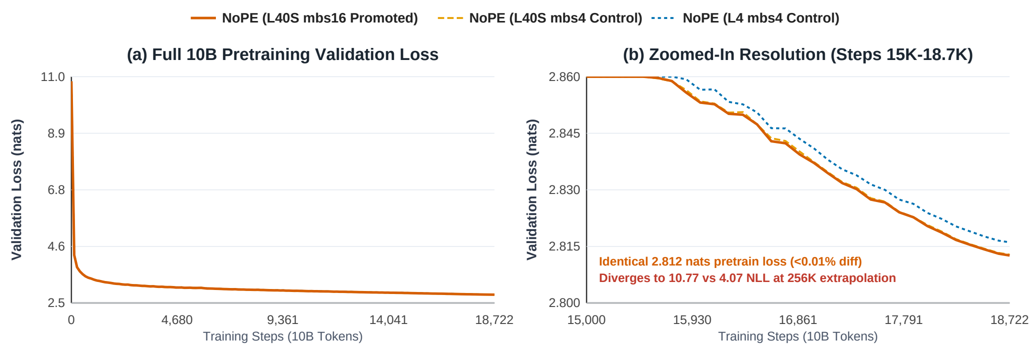

Figure 4: NoPE 2K validation trajectories are nearly identical, yet the checkpoints' 256K NLL values are 10.77, 9.90, and 4.07. The two L40S runs differ in microbatch size. The L4 run also differs in device class.

Figure 4 shows the trigger for a post-hoc audit. Three NoPE runs finish at 2.8126, 2.8128, and 2.8160 validation nats at 2K, yet reach 10.77, 9.90, and 4.07 NLL at 256K. The runs were not initialized from a recorded common seed, so the comparison establishes checkpoint variability, not a device-class treatment effect. Evaluation replays, a microbatch control, and deterministic fixtures ruled out several immediate explanations but did not identify the origin (Appendix K).

6.1 Parameter inspection

We inspected the retention branch rather than whole-block gains. For each layer and query/memory head, we evaluate the learned zero-input operating point $\gamma_0 = \sigma(b_\gamma)$ and convert it to $H_{1/2} = \log(1/2)/\log \gamma_0$. Although the attention path has two KV heads, the memory gate has eight independently learned query/memory-head rows. The promoted NoPE checkpoint contains one block-2 head with a 21.01M-token half-life, and Polar contains one at 3.07M tokens. The corresponding L4 NoPE maximum is 185 tokens. Atma-Raven-Titans stays between 30.5 and 41.4 tokens. Because W_γ is nonzero, runtime retention still depends on activations. The scan is therefore a diagnostic operating point, not a complete trace. The outliers persist after BABILong fine-tuning, at 20.86M tokens for NoPE and 3.05M for Polar.

6.2 Single-head intervention and full re-evaluation

To test causality, we first conducted a paired multi-clamp sweep over candidate caps on the single highest zero-input operating point: checkpoint-local percentiles (p90, p99) and absolute half-life ceilings (hl:256, hl:512) across sequence lengths from 2K to 256K (Table 19). While p99 failed on saturated models because the 99th percentile was still dominated by the outlier ($H_{1/2} \approx 484$k tokens), the 256-token half-life cap

$$\gamma_{\max} = 2^{-1/256} = 0.997296 \tag{5}$$

gave the largest long-distance cross-entropy reduction while keeping 2K clean-loss drift within the 0.05-nat guardrail. In this same-process paired sweep, clean NLL at 256K fell from 12.003 to 1.138 for NoPE and from 1.252 to 1.057 for Polar. Needle cross-entropy fell from 12.20 to 6.05 and from 2.39 to 1.71, respectively. RoPE changed only from 6.16 to 6.02 and lost short-range retrieval, providing an empirical negative control.

We then used the selected 256-token ceiling to re-evaluate downstream tasks, retrieval, fixed-target BPB, and BABILong across the matched attention variants. Figure 1 plots the complete length curves, Table 5 gives the endpoints, and Appendix I provides full numerical tables. All 15 jobs completed without an out-of-memory failure.

Model	Max $H_{1/2}$	Base mean	256K BPB	256K retrieval tok./exact	256K BABI
NoPE	21.01M	45.15 → 45.10	8.097 → 1.595	0.60/0.00 → 0.93/0.00	0 → 39
Polar	3.07M	43.29 → 43.25	1.825 → 1.501	34.37/9.00 → 47.07/16.33	28 → 42
RoPE	682	44.60 → 44.66	2.512 → 2.460	0.00/0.00 → 0.00/0.00	14 → 11

Table 5: Inference-only retention diagnostic. Results show archived baseline → one-head 256-token half-life cap. Base mean averages eight 2K tasks (%). Retrieval is token/exact accuracy (%), BABI is macro exact match, and BPB averages three fixed-target datasets. The cap is a probe, not a proposed trained model.

The intervention largely removes NoPE's likelihood collapse and raises BABILong from 21/6/0/0/0% to 54/45/37/35/39% at 16/32/64/128/256K. However, NoPE's exact natural-text retrieval remains zero beyond 8K and zero overall at 256K. State stability alone is therefore insufficient for exact natural-text recall. For Polar, the cap improves 256K token retrieval by 12.70 points, overall exact retrieval by 7.33 points, and BABILong by 14 points. Exact synthetic retrieval reaches 32.7%, while exact real-text retrieval reaches 91.3% at 8K and remains nonzero through 128K. Mean BPB degradation falls from 1.26× to 1.03×. Mean accuracy on the eight 2K tasks changes by less than 0.06 points. RoPE does not recover retrieval, and its 256K BABILong score falls. The cap is therefore not a generic benchmark boost. Together, these results identify excessive retention as a major checkpoint-specific mediator of NoPE and Polar degradation. The remaining difference between NoPE and Polar is also consistent with a role for Polar's magnitude–direction channel in exact associative recall.

6.3 What this audit cannot answer

The audit leaves several questions unresolved:

1. What training event produces the isolated outlier, and does it replicate under seed-paired device, kernel, and optimizer interventions?
2. Would a bounded train-time parameterization prevent the outlier without removing useful long timescales? The inference cap does not answer this.
3. Why does NoPE exact retrieval remain zero on real text after likelihood and BABILong recover, and why does RoPE not benefit from the same cap?
4. Does Polar's remaining natural-text retrieval gap at 256K improve with longer or mixed-domain training, or does it reflect distractor dispersion under unadapted extrapolation?

These questions delimit the current evidence rather than extend its claims. Resolving them requires interventions that separately vary architecture, optimizer, data, and execution environment.

7 Limitations and Conclusion

This study uses one model scale, one 2K pretraining regime, and too few seed replicates to estimate population-level architectural uncertainty. The Stage I factorial tests robustness across nuisance-factor settings, not random seeds. Teacher-forced retrieval, custom BABI-Long adaptation, and single-sample serving measurements answer different questions and should not be reduced to one score. Untouched Polar reaches zero exact FinePDFs retrieval at 64K–256K, as do the tested baselines at model-dependent lengths. Polar delays this failure but does not solve it. Raven is not an optimizer-matched ablation. The recurrent bound requires a uniform learned margin that the sigmoid parameterization does not enforce. The one-head cap diagnoses existing checkpoints. It neither identifies the outlier's training origin nor validates a bounded train-time formulation. The broader re-evaluation uses pinned archived baselines, while the smaller sweep provides controlled pairing. A new integrated, single-seed component pilot finds that the direction-only path retains most retrieval and supports learned length temperature, but unexpectedly favors a fixed null slope for retrieval; we retain the prespecified Full Polar comparison and leave multi-seed null calibration to future work (Appendix C.2). Our comparisons do not cover every length-scaling method.

Within those limits, ATMA establishes a reproducible operating point for bounded full-context attention plus recurrent memory. Polar materially extends target-token signal, preserves exact synthetic retrieval, and stabilizes long-document likelihood while remaining competitive on short-context tasks. Raven illustrates the BABILong and serving advantages of compressed recurrent state. The checkpoint audit adds a mechanistic qualification: nominally stable gates can sit arbitrarily close to one, and a single extreme retention horizon can dominate extrapolation without appearing in 2K loss. Selective inference-time recovery makes that failure observable and falsifiable while leaving its training origin open. We release configurations, logs, checkpoint manifests, benchmark matrices, clamp specifications, and numerical proof artifacts so that each comparison can be audited.

AI Use Statement

Generative AI tools were used in this revision to assist diagnostic-script implementation, edit and structure the manuscript, and aggregate archived and newly produced evaluation logs. The authors ran the benchmark jobs on the listed checkpoints and hardware. AI was not used to generate evaluation datasets or train the reported models. The authors manually compare all AI-assisted mathematical text with the original derivations and proof artifacts, verify empirical values against source logs, and take responsibility for every claim and artifact in the submission. This statement describes the present version as required by the ICLR 2027 policy.

Reproducibility Statement

Model definitions and training settings are described in Sections 3–4. Appendix D specifies evaluation protocols and denominators, Appendix I records the parameter scan, paired intervention, and full re-evaluation, and Appendices E–J provide detailed tables and diagnostics. The anonymous artifact at https://anonymous.4open.science/r/atma contains configurations, logs, checkpoint manifests, benchmark payloads, clamp specifications, and machine-checked proof skeletons.

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A Theoretical Framing

These are sufficient-condition results. They establish noise-odds and norm bounds under explicit assumptions. They do not establish distributional invariance or replace the empirical evidence.

A.1 Extreme-value null floor

Assumption 1 (Sub-Gaussian noise scores). For irrelevant keys $j \in \mathcal{N}_n$, $|\mathcal{N}_n| \le n$, scores X_j are mean-zero and v^2-sub-Gaussian: $\mathbb{E}\exp(tX_j) \le \exp(v^2t^2/2)$ for all $t \in \mathbb{R}$.

Let the Polar null floor be $\nu_n = b + s\sqrt{\log(n+1)}$ with $b, s \ge 0$.

Proposition 1 (Noise exceedances). Under Assumption 1,

$$\mathbb{E}\left[\sum_{j\in\mathcal{N}_n} \mathbf{1}\{X_j > \nu_n\}\right] \le n\exp\left(-\frac{\nu_n^2}{2v^2}\right). \tag{6}$$

If $s \ge \sqrt{2}v$ this is $O(1)$; if the inequality is strict it decays polynomially.

Proof. Apply the sub-Gaussian tail bound to each key and sum. Since $b \ge 0$, $\nu_n^2 \ge s^2\log(n+1)$. □

Unlike TDA, Polar retains sub-threshold keys. With temperature $\theta_n = 1 + a\log n$, define the unnormalized noise-to-null odds $R_n = \sum_{j\in\mathcal{N}_n}\exp(\theta_n(X_j - \nu_n))$.

Proposition 2 (Vanishing null odds). Suppose $s > c > \sqrt{2}v$ and the null floor is $s\sqrt{\log(n+1)}$ up to a nonnegative offset. With probability tending to one,

$$R_n \le n\exp\Big(-(1 + a\log n)(s-c)\sqrt{\log n}\Big). \tag{7}$$

For $a > 0$, the bound converges to zero faster than any inverse polynomial.

Proof. A union bound gives $\Pr[\max_j X_j > c\sqrt{\log(n+1)}] \le n(n+1)^{-c^2/(2v^2)} \to 0$. On the complementary event, every noise score is at least $(s-c)\sqrt{\log n}$ below the floor. Bound each summand and sum over at most n keys. □

A.2 Bounded channels and recurrent state

Proposition 3 (Bounded Polar readout). For $c_i = s_i / \max(\|s_i\|_2, \varepsilon)$, $\|c_i\|_2 \leq 1$. For $\mu_i = \tanh(\beta \log(1 + m_{\mathrm{eff},i}))$, $\mu_i \in [0, 1)$ when $\beta, m_{\mathrm{eff},i} \geq 0$. A distribution uniform over m real keys has participation ratio m.

Proof. The first two claims follow from the denominator and range of tanh. Uniform weights satisfy $(\sum_{j=1}^{m}(1/m)^2)^{-1} = m$. □

Assumption 2 (Contractive memory). For every t, $\|q_t\|_2 = \|k_t\|_2 = 1$, $\|v_t\|_2 \leq V$, $0 \leq \beta_t \leq 1$, and $0 \leq \gamma_t \leq \bar{\gamma} < 1$.

Proposition 4 (Length-bounded state). For recurrence equation 4,

$$\|M_t\|_F \leq \bar{\gamma}^t \|M_0\|_F + \frac{V}{1 - \bar{\gamma}}. \tag{8}$$

Proof. $I - \beta_t k_t k_t^\top$ has spectral norm at most one and $\|\beta_t v_t k_t^\top\|_F \leq V$. Thus $\|M_t\|_F \leq \bar{\gamma}\|M_{t-1}\|_F + V$; unrolling gives the result. □

The Lean artifact checks the finite-sum, exponential-monotonicity, bounded-channel composition, and invariant-set induction steps. It takes the sub-Gaussian tail, high-probability margin, and one-step contraction as premises. It is therefore a machine-checked proof skeleton, not end-to-end formal verification. The learned Polar conditions $b \geq 0$ and $s > c > \sqrt{2}v$ require an empirical head-wise audit before they can be asserted for a trained checkpoint. Likewise, $\gamma_t = \sigma(W_\gamma x_t + b_\gamma)$ establishes only $0 < \gamma_t < 1$ at any finite logit; it does not supply one sequence-uniform $\bar{\gamma} < 1$. Appendix I shows that the distinction is practically material: individual zero-input operating points can imply million-token half-lives, although that parameter-only scan does not by itself bound activation-conditioned runtime gates.

B Regularization Modes

The factorial grid varies a representation regularizer applied to the model's signature stream. For every non-baseline mode, the sweep uses the same weight $\alpha_{\mathrm{sig}} = 0.01$ and optimizes

$$\mathcal{L} = (1 - \alpha_{\mathrm{sig}})\mathcal{L}_{\mathrm{LM}} + \alpha_{\mathrm{sig}}\mathcal{L}_{\mathrm{reg}} + \lambda_{\mathrm{dist}}\mathcal{L}_{\mathrm{dist}}. \tag{9}$$

The weak and strong modes follow the two SIGReg families introduced by Weak-SIGReg (Akbar, 2026) and LeJEPA's strong SIGReg objective (Balestriero & LeCun, 2025). The discrete and zipfian modes are local variants that combine those ideas with stronger geometric priors. Table 6 defines all five sweep settings.

Mode	Meaning in the sweep
baseline	No signature-stream regularization; equivalently, $\alpha_{\text{sig}} = 0$.
weak	Weak SIGReg: center a random sketch of the token representations and match its covariance to the identity, $\|\text{Cov}(Sx) - I\|_F$. This encourages decorrelated, unit-variance features at low cost.
strong	Strong SIGReg: project representations onto random one-dimensional directions and match their empirical characteristic functions to the standard Gaussian characteristic function over a fixed frequency grid.
discrete	Local variant: normalize each representation to the sphere of radius $\sqrt{d}$, then apply weak covariance matching. This preserves whitening pressure while encouraging fixed-norm, discrete-code-like directions.
zipfian	Local variant: normalize directions for an orthogonality penalty, then match sorted representation magnitudes to a Zipf profile r^{-1}. This biases the signature stream toward heavy-tailed feature use rather than uniform energy allocation.

Table 6: Regularization modes in the 120-cell factorial sweep. Weak and strong are SIGReg-style covariance and distribution-matching objectives; discrete and zipfian are exploratory local variants.

C Polar Attention Implementation

C.1 Design provenance and scope

Before integration, the Polar choices were developed by falsifying simpler alternatives in a standalone synthetic sandbox. Additive soft counts accumulated small per-key leakage with sequence length; thresholds based on a fixed number of standard deviations retained a constant fraction of noise keys; a fixed null logit was eventually overtaken by the maximum noise score; and an unbounded $\log m$ magnitude exposed the output projection to values beyond its training range. These failures motivated participation ratio, the $\sqrt{\log n}$ null floor, and the bounded $\tanh(\beta \log(1 + m))$ map. The standalone prototype was subsequently removed and is not part of the released artifact. We therefore use these tests as design provenance, not as model-scale component ablations or independently reproducible benchmark evidence. The integrated equations, materialized oracle, streaming implementation, and parity tests described below are retained.

C.2 Integrated component pilot

We subsequently ran a matched network-level component study using the promoted Polar+memory recipe. Each 378.22M-parameter model received 1B tokens at length 2K with the same initialization and data seed; evaluation used the same documents, retrieval prompts, and seed. "Fixed null slope" sets only the nonnegative $\sqrt{\log(n + 1)}$ coefficient in Equation 1 near zero; the null base and null vector remain learned. Table 7 reports the prespecified metrics. The clean and junk columns are NLL in nats/token on coherent FinePDFs and the concatenated FineWeb-Edu validation stream, respectively.

Polar variant	Val. NLL	Clean 2K	Clean 64K	Junk 2K	Junk 64K	Needle 2K / 64K (%)
Full Polar	3.1688	2.810	2.047	3.110	3.129	91.25 / 77.50
Direction only	3.1629	2.709	1.982	3.103	3.124	91.25 / 71.25
Constant magnitude	3.1717	2.734	1.924	3.113	3.134	92.50 / 16.25
Fixed null slope	3.1741	2.725	1.938	3.117	3.142	96.25 / 91.25
Fixed temperature gain	3.1695	2.707	1.967	3.111	3.155	93.75 / 16.25

Table 7: Single-seed, 1B-token Polar component attribution. Needle values are teacher-forced five-token accuracy over 16 trials per length. All variants share the same seed, but this design does not estimate variation across training seeds.

Ordinary language-model losses are tightly grouped: validation NLL spans only 0.0112 nats, and every intervention improves clean NLL relative to Full Polar at matched lengths.

Retrieval is more diagnostic. Direction only retains most of the 64K accuracy (71.25% versus 77.50%), whereas constant magnitude and fixed temperature gain each fall to 16.25%. Thus the direction-only path carries most of the retrieval signal, and the learned length temperature is important. An uninformative constant count signal is harmful, but this study does not establish that the count channel is always necessary.

Unexpectedly, fixing the length-dependent null slope near zero gives the best retrieval at every evaluated length and reaches 91.25% at 64K, while slightly worsening validation and junk-stream NLL. One plausible explanation is that a slope learned only through 2K over-rejects distant relevant keys when extrapolated, especially because this recipe disables the explicit distractor-alignment objective. The matched seed makes an independently luckier initialization an unlikely explanation for this contrast, but one training seed and 16 retrieval trials per length cannot establish a population-level effect. We therefore retain the prespecified Full Polar model for the confirmatory comparison and treat null calibration as unresolved. Future work should replicate the contrast across seeds and larger retrieval samples, measure null mass by head versus length, and test bounded or score-adaptive null formulations. The synthetic provenance above and this integrated pilot answer different questions: the former motivates the asymptotic design, while the latter shows that its learned finite-training calibration is not yet empirically optimal.

C.3 Numerical oracle

The materialized PyTorch path forms the masked real-key logits and virtual null logit in FP32, applies a single softmax, and computes the direction and magnitude channels from the resulting weights. It is used as a numerical oracle for forward and backward tests, not as the production long-context path, because materializing the $T \times T$ score matrix is memory-prohibitive at the evaluated lengths.

C.4 Streaming sufficient statistics

The Triton kernel follows the online-max strategy used by FlashAttention-style kernels. For one query row, it streams K, V blocks and retains only four FP32 statistics: the running maximum M, real-key exponential sum L, squared exponential sum Q_2, and unnormalized value sum S. If a new block changes the row maximum, $\alpha = \exp(M_{\text{old}} - M_{\text{new}})$ rescales each statistic as shown in Table 8.

State	Online update	Use after streaming
M	$M \leftarrow \max(M, \max a)$	Stable common logit origin
L	$L \leftarrow \alpha L + \sum_j p_j$	Real-key softmax mass
Q_2	$Q_2 \leftarrow \alpha^2 Q_2 + \sum_j p_j^2$	$n_{\text{eff}} = L^2/Q_2$
S	$S \leftarrow \alpha S + \sum_j p_j v_j$	Direction numerator

Table 8: Streaming state for one Polar query row, where a is the current logit block and $p_j = \exp(a_j - M_{\text{new}})$. The squared accumulator must rescale by α^2.

After the real keys are consumed, the virtual null is folded in as one additional softmax entry. Let $p_0 = \exp(t_i \nu_i - M)$ and $Z = L + p_0$. The kernel obtains

$$c_i = \text{normalize}(S + p_0 v_{\text{null}}), \qquad n_{\text{eff},i} = \frac{L^2}{Q_2}, \qquad m_{\text{eff},i} = n_{\text{eff},i}\frac{L}{Z}. \tag{10}$$

The common factor $1/Z$ cancels under radial normalization of the direction numerator. The production kernel evaluates the magnitude with $2\sigma(2x) - 1 = \tanh x$ and saves (M, L, Q_2, S) for the backward pass. Listing 1 gives the algorithmic core; pointer arithmetic, launch metadata, and backward code are omitted.

Listing 1: Schematic core of the block-streamed Triton forward pass.

```
@triton.jit
def polar_fwd_core(q, K, V, n_i, temp, nullv, beta,
```

```
                   BLOCK_N: tl.constexpr, DK: tl.constexpr):
    M  = tl.full([BLOCK_M], -1e38, tl.float32)
    L  = tl.zeros([BLOCK_M], tl.float32)
    Q2 = tl.zeros([BLOCK_M], tl.float32)
    S  = tl.zeros([BLOCK_M, DK], tl.float32)

    for start in range(0, n_i, BLOCK_N):
        k, v, valid = load_kv_block(K, V, start, n_i)
        a = tl.where(valid, temp[:, None] * dot(q, k), -1e38)
        M_new = tl.maximum(M, tl.max(a, axis=1))
        alpha = tl.exp(M - M_new)
        p = tl.where(valid, tl.exp(a - M_new[:, None]), 0.0)
        L  = alpha * L + tl.sum(p, axis=1)
        Q2 = alpha * alpha * Q2 + tl.sum(p * p, axis=1)
        S  = alpha[:, None] * S + tl.dot(p, v)
        M = M_new

    # Fold in the learned virtual null as one extra entry.
    M_new = tl.maximum(M, temp * nullv)
    alpha = tl.exp(M - M_new)
    L, Q2, S = alpha * L, alpha * alpha * Q2, alpha[:, None] * S
    p_null = tl.exp(temp * nullv - M_new)
    Z = L + p_null

    direction = normalize(S + p_null[:, None] * v_null)
    n_eff = L * L / tl.maximum(Q2, eps)
    m_eff = n_eff * L / tl.maximum(Z, eps)
    magnitude = 2 * tl.sigmoid(2 * beta * tl.log(1 + m_eff)) - 1
    save_for_backward(M_new, L, Q2, S)
```

This removes the quadratic score-matrix storage. Full-context prefill still performs quadratic query-key arithmetic. The kernel is cross-checked against the materialized PyTorch oracle in FP32 accumulation before lower-precision outputs are accepted.

D Evaluation Protocols

Stage I contains 120 approximately 370–378M-parameter runs trained for 1,900 steps of 524,288 tokens at sequence length 2,048 on FineWeb-Edu. Stage II contains three matched attention models trained for 18,722 such steps (9.816B tokens). The five promoted checkpoints and ten open-weight controls are evaluated with checkpoint-exact forward paths. Stage I cells vary factorial settings and are not independent seed replicates.

D.1 Raven optimizer control

The Raven rows use AdamW optimizer because the attempted matched-schedule control did not produce a usable baseline. In the 1B-token Raven Native pilot with the Atma Muon schedule, validation loss decreased from 10.83 to 3.76 nats by step 625, then rose to 6.88 at step 1,125 and ended at 4.62; teacher-forced retrieval was 0% at every tested length from 2K through 64K. The reported Raven Native and Atma-Raven-Titans checkpoints therefore use AdamW and remain a separate model-family/optimizer group. This is evidence about the tested Muon schedule, not a general claim that Raven cannot converge under any Muon tuning.

Retrieval crosses passkey and NIAH tasks, synthetic and FinePDFs haystacks, eight lengths from 2K through 256K, three depths, and 50 paired trials per cell. The scorer appends the complete five-token target and evaluates each position conditioned on the preceding ground-truth target tokens. Token accuracy is the fraction of correct per-position argmax predictions; exact accuracy requires all five positions to be correct. Both are teacher-forced diagnostics, not autoregressive generation. An older free-generation protocol was retired because all base checkpoints scored uniformly zero. BABILong uses answer-only fine-tuning on tasks qa1–qa10 at 0K/1K/2K and greedy exact match on ten held-out rows per task and length. Base-task controls use LAMBADA, HellaSwag, PIQA, WinoGrande, ARC-Easy, ARC-Challenge, OpenBookQA, and BoolQ. Fixed-target BPB uses FinePDFs, PG-19, and Proof-Pile. All reported processes completed without failed, OOM, unsupported, missing, or null cells.

For proportions we report simple Wilson intervals where they aid interpretation. These intervals treat trials as binomial observations and do not exploit the paired design or account for correlations induced by shared templates. The archived retrieval payloads retain aggregate cell counts but the BABILong payloads do not retain per-model correctness vectors, so paired bootstrap or McNemar tests cannot be reconstructed without rerunning or recovering the original predictions.

E Complete Factorial Sweep

Table 9 retains all 120 factorial cells; the main paper reports only the recipe transitions needed to justify promotion.

Table 9: Complete Results of the 120-Run Ablation Sweep. All models are 370M parameters, trained on FineWeb-Edu for 1B tokens ($seq\_len = 2048$). Evaluated from 2K to 64K context length. Clean NLL is measured on coherent FinePDFs documents; junk NLL is measured on the concatenated validation stream. NLL is in nats/token (lower is better), Needle is greedy per-digit accuracy in % (higher is better), and MFU is training model flops utilization in %.

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	baseline	Off	Off	Off	3.323	2.83	3.60	3.27	5.68	96.3	0.0	40.1
POLAR	baseline	Off	Off	On	3.343	2.83	2.66	3.30	4.74	67.5	0.0	39.6
POLAR	baseline	Off	On	Off	3.169	2.70	1.96	3.11	3.13	91.3	92.5	35.6
POLAR	baseline	Off	On	On	3.174	2.71	2.01	3.12	3.13	51.3	30.0	35.6
POLAR	baseline	On	Off	Off	3.332	2.81	5.56	3.28	6.54	85.0	0.0	33.3
POLAR	baseline	On	Off	On	3.361	2.85	2.89	3.35	4.97	91.3	2.5	34.5
POLAR	baseline	On	On	Off	3.178	2.72	1.98	3.12	3.14	73.8	58.8	31.8
POLAR	baseline	On	On	On	3.182	2.73	1.97	3.12	3.14	56.3	21.3	31.2
POLAR	weak	Off	Off	Off	3.333	2.82	2.86	3.28	5.35	88.8	0.0	39.0
POLAR	weak	Off	Off	On	3.348	2.90	3.76	3.38	5.24	93.8	0.0	37.9
POLAR	weak	Off	On	Off	3.179	2.74	1.95	3.12	3.14	70.0	51.3	36.2
POLAR	weak	Off	On	On	3.187	2.74	1.99	3.13	3.15	41.3	41.3	36.2
POLAR	weak	On	Off	Off	3.335	2.82	3.47	3.28	5.56	91.3	0.0	32.8
POLAR	weak	On	Off	On	3.353	2.82	2.58	3.32	4.81	96.3	1.3	32.8
POLAR	weak	On	On	Off	3.178	2.74	1.95	3.12	3.14	33.8	32.5	30.7
POLAR	weak	On	On	On	3.183	2.74	1.94	3.13	3.15	81.3	78.8	30.7
POLAR	strong	Off	Off	Off	3.338	2.85	3.79	3.28	5.83	98.8	1.3	37.3
POLAR	strong	Off	Off	On	3.360	2.85	2.99	3.32	5.01	91.3	0.0	37.9
POLAR	strong	Off	On	Off	3.172	2.72	1.96	3.11	3.13	80.0	52.5	35.8
POLAR	strong	Off	On	On	3.181	2.72	1.94	3.12	3.15	76.3	68.8	34.6

Continued on next page

Table 9: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
POLAR	strong	On	Off	Off	3.337	2.87	3.22	3.28	5.52	83.8	1.3	32.8
POLAR	strong	On	Off	On	3.347	2.82	2.64	3.30	4.88	97.5	16.3	32.2
POLAR	strong	On	On	Off	3.179	2.73	1.94	3.12	3.14	86.3	83.8	30.9
POLAR	strong	On	On	On	3.182	2.72	1.94	3.12	3.14	25.0	23.8	30.9
POLAR	discrete	Off	Off	Off	3.338	2.82	2.80	3.29	5.12	93.8	2.5	37.9
POLAR	discrete	Off	Off	On	3.349	2.87	2.90	3.33	5.25	93.8	6.3	37.9
POLAR	discrete	Off	On	Off	3.184	2.75	2.03	3.12	3.14	76.3	73.8	36.4
POLAR	discrete	Off	On	On	3.178	2.71	1.92	3.12	3.15	31.3	18.8	35.8
POLAR	discrete	On	Off	Off	3.333	2.80	4.13	3.28	6.12	97.5	0.0	32.6
POLAR	discrete	On	Off	On	3.351	2.87	2.72	3.33	4.89	85.0	2.5	33.3
POLAR	discrete	On	On	Off	3.189	2.74	1.96	3.13	3.15	87.5	53.8	31.8
POLAR	discrete	On	On	On	3.192	2.74	2.07	3.13	3.15	21.3	26.3	31.7
POLAR	zipfian	Off	Off	Off	3.338	2.85	4.83	3.28	6.55	96.3	11.3	37.9
POLAR	zipfian	Off	Off	On	3.350	2.83	2.58	3.32	4.63	88.8	3.8	39.4
POLAR	zipfian	Off	On	Off	3.171	2.70	1.92	3.11	3.13	52.5	63.8	36.2
POLAR	zipfian	Off	On	On	3.180	2.75	1.95	3.12	3.15	46.3	40.0	35.2
POLAR	zipfian	On	Off	Off	3.340	2.83	3.77	3.28	5.64	98.8	1.3	33.8
POLAR	zipfian	On	Off	On	3.352	2.85	3.13	3.36	5.23	90.0	0.0	33.5
POLAR	zipfian	On	On	Off	3.177	2.72	1.94	3.12	3.14	91.3	67.5	31.8
POLAR	zipfian	On	On	On	3.192	2.74	2.00	3.13	3.15	40.0	26.3	31.8
NOPE	baseline	Off	Off	Off	3.224	2.76	3.71	3.16	6.68	97.5	1.3	41.5
NOPE	baseline	Off	Off	On	3.219	2.75	2.76	3.17	5.49	92.5	10.0	36.5
NOPE	baseline	Off	On	Off	3.140	2.66	2.34	3.09	3.23	97.5	16.3	36.8
NOPE	baseline	Off	On	On	3.150	2.70	2.14	3.09	3.24	81.3	0.0	34.2
NOPE	baseline	On	Off	Off	3.209	2.74	2.99	3.15	6.13	90.0	18.8	32.5
NOPE	baseline	On	Off	On	3.212	2.77	2.90	3.17	5.65	81.3	0.0	30.3
NOPE	baseline	On	On	Off	3.149	2.67	2.35	3.10	3.28	80.0	16.3	30.0
NOPE	baseline	On	On	On	3.147	2.68	2.09	3.09	3.23	88.8	21.3	28.2
NOPE	weak	Off	Off	Off	3.214	2.77	3.10	3.16	6.34	95.0	0.0	40.9
NOPE	weak	Off	Off	On	3.222	2.75	3.05	3.19	5.81	81.3	0.0	36.1

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Table 9: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
NOPE	weak	Off	On	Off	3.147	2.68	2.07	3.09	3.19	88.8	21.3	37.5
NOPE	weak	Off	On	On	3.142	2.68	2.12	3.08	3.23	82.5	0.0	33.7
NOPE	weak	On	Off	Off	3.215	2.75	4.87	3.16	12.48	95.0	5.0	32.0
NOPE	weak	On	Off	On	3.221	2.73	2.63	3.18	4.84	81.3	0.0	29.3
NOPE	weak	On	On	Off	3.147	2.67	2.48	3.09	3.23	97.5	1.3	30.8
NOPE	weak	On	On	On	3.148	2.68	2.14	3.09	3.24	93.8	1.3	28.4
NOPE	strong	Off	Off	Off	3.231	2.74	4.88	3.18	11.85	90.0	2.5	38.4
NOPE	strong	Off	Off	On	3.214	2.72	2.68	3.17	5.28	82.5	0.0	36.4
NOPE	strong	Off	On	Off	3.144	2.68	2.37	3.09	3.24	96.3	6.3	35.7
NOPE	strong	Off	On	On	3.150	2.70	2.10	3.10	3.25	93.8	7.5	33.8
NOPE	strong	On	Off	Off	3.207	2.72	3.13	3.15	6.20	73.8	0.0	31.4
NOPE	strong	On	Off	On	3.214	2.74	2.49	3.17	4.36	86.3	8.8	28.9
NOPE	strong	On	On	Off	3.146	2.67	2.24	3.09	3.23	83.8	17.5	29.1
NOPE	strong	On	On	On	3.147	2.68	2.16	3.09	3.28	96.3	0.0	27.5
NOPE	discrete	Off	Off	Off	3.202	2.75	3.33	3.15	6.23	85.0	1.3	38.8
NOPE	discrete	Off	Off	On	3.234	2.79	3.11	3.19	5.47	70.0	18.8	37.0
NOPE	discrete	Off	On	Off	3.145	2.68	2.14	3.09	3.27	98.8	16.3	36.0
NOPE	discrete	Off	On	On	3.145	2.68	2.10	3.09	3.25	95.0	3.8	34.3
NOPE	discrete	On	Off	Off	3.212	2.88	3.04	3.16	6.88	42.5	0.0	31.2
NOPE	discrete	On	Off	On	3.211	2.78	2.73	3.16	5.28	82.5	1.3	29.1
NOPE	discrete	On	On	Off	3.145	2.68	2.27	3.08	3.26	91.3	11.3	30.3
NOPE	discrete	On	On	On	3.149	2.68	2.17	3.08	3.32	83.8	0.0	28.6
NOPE	zipfian	Off	Off	Off	3.209	2.71	3.18	3.15	6.40	90.0	0.0	40.8
NOPE	zipfian	Off	Off	On	3.219	2.75	2.71	3.17	5.44	77.5	1.3	36.1
NOPE	zipfian	Off	On	Off	3.141	2.67	2.55	3.08	3.38	96.3	0.0	38.0
NOPE	zipfian	Off	On	On	3.150	2.69	2.11	3.09	3.25	90.0	3.8	33.7
NOPE	zipfian	On	Off	Off	3.207	2.77	3.32	3.15	7.09	82.5	0.0	31.3
NOPE	zipfian	On	Off	On	3.227	2.79	2.91	3.18	5.94	93.8	11.3	30.2
NOPE	zipfian	On	On	Off	3.146	2.68	2.39	3.10	3.25	96.3	21.3	29.6
NOPE	zipfian	On	On	On	3.148	2.68	2.17	3.09	3.28	90.0	6.3	28.7

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Table 9: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
ROPE	baseline	Off	Off	Off	3.159	2.85	3.36	3.09	5.21	42.5	0.0	40.4
ROPE	baseline	Off	Off	On	3.163	2.86	3.79	3.17	5.86	15.0	0.0	37.2
ROPE	baseline	Off	On	Off	3.145	2.81	2.86	3.09	3.57	73.8	0.0	37.1
ROPE	baseline	Off	On	On	3.151	2.80	2.95	3.12	3.72	13.8	0.0	34.4
ROPE	baseline	On	Off	Off	3.163	3.00	3.47	3.10	5.46	38.8	0.0	33.2
ROPE	baseline	On	Off	On	3.167	2.88	4.06	3.16	5.92	0.0	0.0	30.2
ROPE	baseline	On	On	Off	3.153	2.73	2.45	3.09	3.57	75.0	0.0	31.4
ROPE	baseline	On	On	On	3.154	2.83	2.86	3.12	3.72	32.5	0.0	29.4
ROPE	weak	Off	Off	Off	3.158	2.88	3.46	3.10	5.36	46.3	0.0	41.4
ROPE	weak	Off	Off	On	3.163	2.84	3.96	3.18	5.93	12.5	0.0	37.8
ROPE	weak	Off	On	Off	3.149	2.71	2.46	3.09	3.56	37.5	0.0	36.5
ROPE	weak	Off	On	On	3.156	2.85	2.77	3.12	3.60	23.8	0.0	34.9
ROPE	weak	On	Off	Off	3.155	2.83	3.49	3.10	5.27	38.8	0.0	32.3
ROPE	weak	On	Off	On	3.157	3.05	4.12	3.18	5.86	8.8	0.0	30.5
ROPE	weak	On	On	Off	3.149	2.77	2.76	3.09	3.56	57.5	0.0	31.1
ROPE	weak	On	On	On	3.149	2.77	3.13	3.12	3.80	17.5	0.0	28.0
ROPE	strong	Off	Off	Off	3.165	2.78	3.34	3.10	5.27	30.0	0.0	39.0
ROPE	strong	Off	Off	On	3.169	2.90	3.68	3.19	5.80	31.3	0.0	36.3
ROPE	strong	Off	On	Off	3.158	2.85	2.63	3.10	3.50	51.3	0.0	35.9
ROPE	strong	Off	On	On	3.162	2.86	3.08	3.14	3.76	10.0	0.0	33.2
ROPE	strong	On	Off	Off	3.161	2.85	3.58	3.10	5.28	47.5	0.0	31.1
ROPE	strong	On	Off	On	3.165	2.85	4.03	3.19	5.90	2.5	0.0	29.2
ROPE	strong	On	On	Off	3.155	2.80	2.70	3.10	3.62	61.3	0.0	29.3
ROPE	strong	On	On	On	3.162	2.85	3.08	3.13	3.74	2.5	0.0	28.4
ROPE	discrete	Off	Off	Off	3.165	2.91	3.43	3.11	5.42	22.5	0.0	39.7
ROPE	discrete	Off	Off	On	3.158	2.84	4.05	3.15	5.90	3.8	0.0	36.9
ROPE	discrete	Off	On	Off	3.159	2.83	2.67	3.11	3.56	47.5	0.0	36.3
ROPE	discrete	Off	On	On	3.161	2.79	2.71	3.13	3.74	41.3	0.0	33.7
ROPE	discrete	On	Off	Off	3.171	2.79	3.45	3.12	5.36	51.3	0.0	31.5
ROPE	discrete	On	Off	On	3.164	2.80	3.74	3.16	5.86	21.3	0.0	30.0

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Table 9: Complete Results of the 120-Run Ablation Sweep (continued).

Type	Reg	Dist	Mem	Win	Val	Clean2K	Clean64K	Junk2K	Junk64K	Ndl2K	Ndl64K	MFU
ROPE	discrete	On	On	Off	3.162	2.74	2.47	3.11	3.53	50.0	0.0	30.9
ROPE	discrete	On	On	On	3.154	2.79	2.68	3.11	3.67	18.8	0.0	28.9
ROPE	zipfian	Off	Off	Off	3.163	2.82	3.25	3.10	5.22	62.5	0.0	39.8
ROPE	zipfian	Off	Off	On	3.161	2.94	3.94	3.18	5.81	16.3	0.0	37.1
ROPE	zipfian	Off	On	Off	3.149	2.78	2.70	3.09	3.63	46.3	0.0	36.7
ROPE	zipfian	Off	On	On	3.151	2.80	2.63	3.11	3.68	2.5	0.0	33.9
ROPE	zipfian	On	Off	Off	3.161	2.78	3.38	3.10	5.32	47.5	0.0	32.6
ROPE	zipfian	On	Off	On	3.160	3.01	4.11	3.17	5.81	17.5	0.0	30.2
ROPE	zipfian	On	On	Off	3.149	2.73	2.39	3.09	3.55	62.5	0.0	29.8
ROPE	zipfian	On	On	On	3.152	2.79	2.99	3.12	3.78	26.3	0.0	28.5

F Retrieval Breakdown

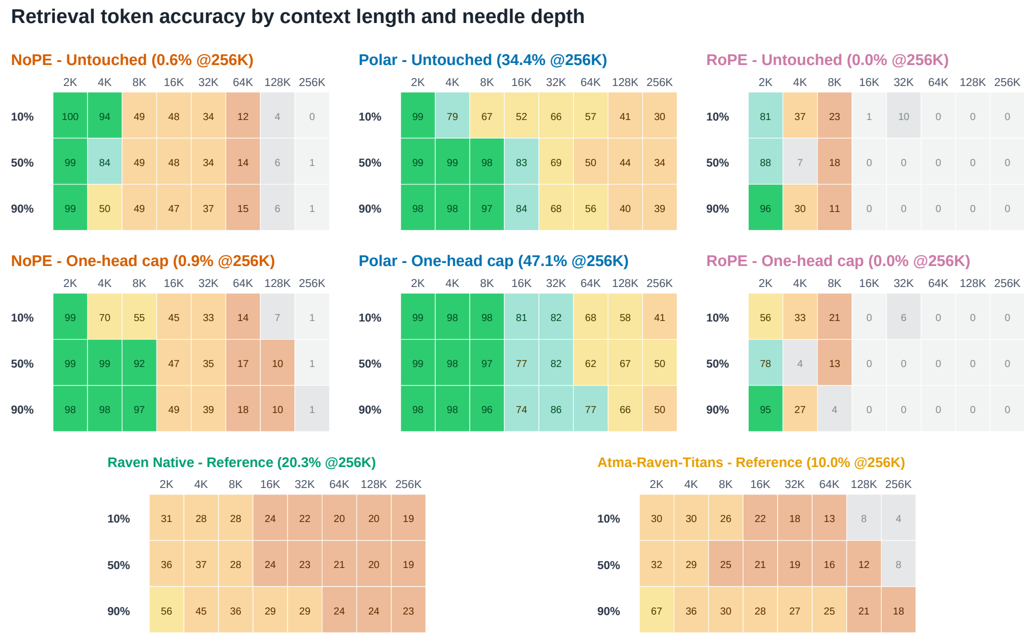

Figure 5: Retrieval depth sweeps for the three matched attention variants before and after the one-head retention cap, followed by untouched Raven Native and Atma-Raven-Titans reference panels. Each cell is teacher-forced token accuracy averaged over passkey and NIAH and over synthetic and FinePDFs haystacks. The Raven rows are separately optimized comparisons, not clamped ablations.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	99.6	98.4	98.5	97.0	94.9	92.3	80.3	68.4
	FinePDFs	98.6	86.3	77.9	49.1	40.7	16.3	3.2	0.3
NoPE	Synthetic	99.5	98.7	98.9	94.9	70.3	26.5	10.2	1.2
	FinePDFs	99.3	52.9	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	81.7	44.9	33.1	0.8	6.8	0.1	0.0	0.0
	FinePDFs	94.8	4.7	0.2	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	64.8	60.7	51.1	46.9	41.7	35.8	25.9	19.3
	FinePDFs	23.2	2.6	0.7	0.2	1.4	0.5	1.6	0.7
Raven Native	Synthetic	55.9	57.7	51.4	48.3	44.3	40.9	38.9	35.1
	FinePDFs	22.1	15.4	8.2	3.3	5.3	2.9	3.7	5.4

Table 10: Zero-shot retrieval accuracy (%) by haystack type and context length, averaged over needle depths.

Figure 5 shows variation by needle depth across all five models. Tables 10 and 11 separate partial target-token signal from exact five-token recall. Every tested model eventually reaches 0.0% exact retrieval on real-text haystacks under unadapted length extrapolation:

- Raven Native and Atma-Raven-Titans: Raven Native scores 0.0% on FinePDFs at every tested length and no more than 1.0% on synthetic haystacks. Atma-Raven-Titans scores 4.7% at 2K on FinePDFs and 0.0% from 4K onward.
- RoPE: Exact FinePDFs retrieval falls from 77.3% at 2K to 0.0% at 4K and stays there through 256K.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
Polar	Synthetic	98.2	92.3	93.2	86.0	76.0	65.7	26.3	18.0
	FinePDFs	93.0	67.3	62.7	12.0	0.0	0.0	0.0	0.0
NoPE	Synthetic	97.7	93.7	94.3	78.0	23.7	4.7	0.7	0.0
	FinePDFs	96.7	18.0	0.0	0.0	0.0	0.0	0.0	0.0
RoPE	Synthetic	42.0	4.3	0.3	0.0	0.0	0.0	0.0	0.0
	FinePDFs	77.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Atma-Raven-Titans	Synthetic	20.7	5.7	1.7	2.0	0.7	0.3	0.0	0.0
	FinePDFs	4.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Raven Native	Synthetic	1.0	1.0	0.2	0.0	0.0	0.0	0.0	0.0
	FinePDFs	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 11: Zero-shot exact five-token retrieval accuracy (%) by haystack type and context length, averaged over needle depths. On FinePDFs, Raven Native remains at 0.0% across all lengths, RoPE reaches 0.0% at 4K, and NoPE reaches 0.0% at 8K. Polar retains 67.3% at 4K, 62.7% at 8K, and 12.0% at 16K before reaching 0.0% at longer contexts.

- NoPE: Exact FinePDFs retrieval falls from 96.7% at 2K to 18.0% at 4K and 0.0% from 8K onward.
- Polar Attention: Untouched Polar reaches 93.0%, 67.3%, 62.7%, and 12.0% on FinePDFs at 2K, 4K, 8K, and 16K. On synthetic haystacks, it retains 65.7% exact retrieval at 64K and 18.0% at 256K, with a 95% Wilson interval of 14.1–22.7% at 256K. The other models score 0.0% there.

Thus, the loss of exact natural-text retrieval at long context is not unique to ATMA. Polar delays this loss relative to the tested baselines but does not eliminate it.

F.1 Open-weight context

Table 12 reports the ten archived open-weight controls. They differ in training data, tokenizer, architecture, parameter count, and native context length; several were trained at lengths beyond 2K. They are therefore practical context, not matched length-extrapolation baselines and not evidence of a Polar advantage over the corresponding model families.

Open-weight model	NLL 2K	NLL 64K	Needle 2K	Needle 64K
SmolLM2-360M	1.99	6.04	100.0	8.1
LFM2.5-230M	3.01	6.69	38.1	19.4
LFM2.5-350M	3.18	6.63	45.6	46.3
Qwen2.5-0.5B	2.03	1.53	100.0	38.1
Qwen3-0.6B-Base	1.95	1.40	100.0	95.0
Qwen3.5-0.8B-Base	1.94	1.33	100.0	100.0
Gemma-3-270M	2.48	1.90	96.2	24.4
Granite-4.0-350M	2.52	3.02	40.6	8.8
Granite-4.0-H-350M	2.22	2.06	75.0	54.4
Falcon-H1-0.5B	1.99	2.54	92.5	19.4

Table 12: Open-weight controls evaluated through 64K. NLL is fixed-target FinePDFs nats/token; Needle is teacher-forced target-token accuracy over 16 trials. These checkpoints are not matched for data, tokenizer, native context, or training budget.

G Short-Context Quality and Systems

Table 13 gives the exact per-task controls shown in main-text Figure 3. Figure 6 shows the serving-length trend, and Table 14 summarizes the quality–systems trade-off.

Group	Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
Matched	NoPE	30.16	37.88	66.27	51.93	50.53	32.44	31.60	60.37	45.15
Matched	RoPE	30.25	37.19	66.76	50.67	49.12	29.43	32.40	60.95	44.60
Matched	Polar	28.47	36.19	66.87	51.62	49.12	26.09	31.80	56.21	43.29
Raven/AdamW	Atma-Raven-Titans	26.00	34.12	65.56	52.49	48.07	27.76	29.00	61.41	43.05
Raven/AdamW	Raven Native	27.48	33.56	64.69	51.70	49.12	27.09	28.80	61.93	43.05

Table 13: Task-level zero-shot accuracy (%) on the eight short-context controls (22,939 examples per model). LAMBADA, WinoGrande, and BoolQ use standard accuracy; HellaSwag, PIQA, ARC-Easy, ARC-Challenge, and OpenBookQA use length-normalized accuracy. Mean is the unweighted average of the eight displayed metrics; bold is best within each comparison group.

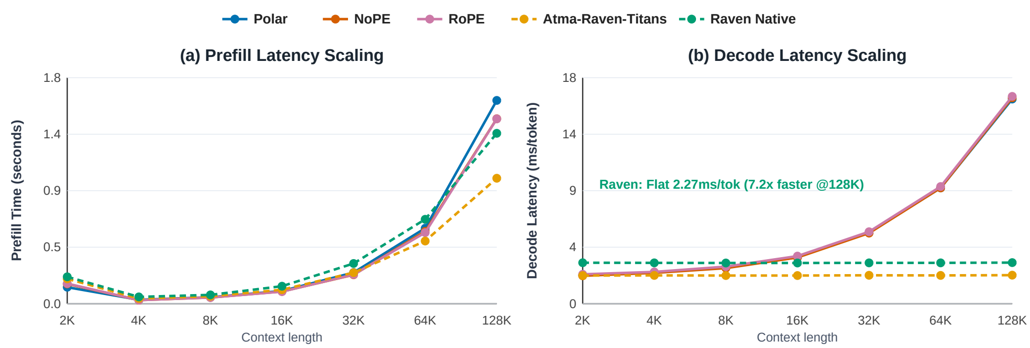

Figure 6: Single-sequence serving on one L40S. Attention decode scales with KV-cache length; recurrent Raven uses fixed state.

H Systems Implementation and Optimization Breakdown

The inference stack uses separate kernels for full-context prefill, paged autoregressive decode, and recurrent-state updates. Table 15 distinguishes what each optimization changes. In particular, streaming removes quadratic score storage but not quadratic prefill arithmetic, while the recurrent Raven path changes the state representation itself and therefore has different scaling.

Group	Model	Base mean (%)	128K prefill (s)	Decode (ms/token)	Peak alloc. (GiB)	Train MFU (%, 2K)
Matched	NoPE	45.1	1.47	16.4	39.40	42.77
Matched	Polar	43.3	1.62	16.3	39.41	40.44
Matched	RoPE	44.6	1.47	16.5	39.41	42.45
Raven/AdamW	Atma-Raven-Titans	43.1	1.00	2.27	8.62	30.92
Raven/AdamW	Raven Native	43.0	1.36	3.27	6.46	21.53

Table 14: Short-context quality and serving trade-offs. Serving uses one sequence and one timing sample; it is descriptive rather than an uncertainty estimate.

Path	Optimization	Operational effect
Polar prefill	Block-streamed K, V tiles with online (M, L, Q_2, S) accumulation	Avoids materializing $T \times T$ scores; retains full-context $O(T^2)$ arithmetic.
Polar decode	Paged KV-cache kernel grouping the four query heads that share each KV head	Reuses KV loads across the GQA group; decode remains $O(T)$ in cache length.
Gated-delta step	Fused in-place decay, prediction, delta write, and readout	Avoids materializing per-step intermediate states and preserves $O(1)$ recurrent state size.
Ragged prefill	Packed prompts with grouped compatible tile shapes	Reduces padding work on heterogeneous prompt batches.
Compiler boundary	Recurrent scan and custom kernels exposed as opaque operations	Prevents compiler graph breaks around the stateful update while retaining explicit numerical tests.

Table 15: Implementation-level optimization breakdown. Complexity statements describe the sequence-mixing path and exclude shared MLP and embedding costs.

H.1 Single-sequence scaling

Table 16 gives the underlying single-sequence measurements. On one NVIDIA L40S, attention decode increases from approximately 2.3 ms/token at 2K to 16.3–16.5 ms/token at 128K because every step reads a longer KV cache. The recurrent models remain nearly flat because they update fixed-size state. Prefill is less monotonic at short lengths because launch overhead and kernel occupancy dominate before the matrix operations reach their efficient regime. These timings use batch size 1, 32 generated tokens, and one recorded timing sample per cell. They are descriptive measurements, not confidence intervals.

Table 16: Detailed prefill and decode scaling from 2K to 128K tokens on one NVIDIA L40S GPU (batch size 1, 32 generated tokens).

Model	Metric	2K	4K	8K	16K	32K	64K	128K
Polar	Prefill latency (s)	0.131	0.031	0.051	0.101	0.248	0.601	1.619
	Prefill throughput (tok/s)	15,640	133,884	160,145	161,551	132,091	109,050	80,959
	Decode latency (ms/tok)	2.27	2.49	2.88	3.70	5.66	9.22	16.30
	Decode throughput (tok/s)	441.2	402.3	346.6	270.1	176.8	108.5	61.3
NoPE	Prefill latency (s)	0.157	0.028	0.049	0.097	0.230	0.576	1.473
	Prefill throughput (tok/s)	13,070	146,568	166,504	168,510	142,319	113,810	89,013
	Decode latency (ms/tok)	2.22	2.45	2.83	3.68	5.62	9.22	16.40
	Decode throughput (tok/s)	450.1	408.8	353.0	271.7	177.8	108.4	61.0
RoPE	Prefill latency (s)	0.160	0.028	0.049	0.096	0.232	0.566	1.472
	Prefill throughput (tok/s)	12,761	147,089	168,287	170,936	141,249	115,786	89,015
	Decode latency (ms/tok)	2.34	2.53	2.95	3.79	5.73	9.33	16.51
	Decode throughput (tok/s)	427.4	395.3	338.7	263.7	174.5	107.2	60.6
Atma-Raven-Titans	Prefill latency (s)	0.198	0.041	0.059	0.111	0.247	0.499	0.999
	Prefill throughput (tok/s)	10,334	100,989	137,903	147,434	132,916	131,464	131,195
	Decode latency (ms/tok)	2.24	2.25	2.24	2.24	2.26	2.25	2.27
	Decode throughput (tok/s)	446.6	445.1	446.0	447.1	442.6	444.8	440.3
Raven Native	Prefill latency (s)	0.214	0.054	0.070	0.139	0.320	0.672	1.357
	Prefill throughput (tok/s)	9,558	75,738	116,378	117,605	102,354	97,527	96,559
	Decode latency (ms/tok)	3.26	3.25	3.24	3.24	3.25	3.25	3.27
	Decode throughput (tok/s)	306.4	307.5	308.2	308.3	307.6	307.5	306.1

H.2 Production-framework stress comparison

To test whether the custom kernels introduce an obvious systems bottleneck at a larger shape, we configured ATMA's inference engine with a 9.2B-parameter stress shape and compared it with vLLM 0.25.0 serving Qwen3.5-9B on the same L40S in BF16 (Kwon et al., 2023). The models are not weight- or architecture-matched, so Table 17 is a backend stress comparison rather than a model-quality or universal serving claim. ATMA trails on short homogeneous prefill, is similar on long heterogeneous prefill, and leads the recorded high-batch decode cells.

Phase	Batch/workload	ATMA engine	vLLM/Qwen3.5-9B	Ratio
Prefill	Homogeneous $B = 8, T = 512$ (4,096 tok)	11,947 tok/s	12,772 tok/s	0.94×
	Heterogeneous short-heavy (816 tok)	5,752 tok/s	8,002 tok/s	0.72×
	Heterogeneous mixed (4,384 tok)	12,779 tok/s	11,996 tok/s	1.07×
	Heterogeneous long-tail (8,192 tok)	12,251 tok/s	11,721 tok/s	1.05×
Decode, context 512	Batch 128	2,939 tok/s	2,247 tok/s	1.31×
	Batch 256	4,501 tok/s	2,921 tok/s	1.54×
	Batch 512	5,919 tok/s	2,558 tok/s	2.31×
	Heterogeneous $B = 128$ (context 64–1536)	2,950 tok/s	2,403 tok/s	1.23×

Table 17: Recorded systems-scale throughput for ATMA's 9.2B stress shape and vLLM 0.25.0 serving Qwen3.5-9B on one L40S (BF16, greedy sampling). Ratio is ATMA/vLLM.

I Retention-Horizon Diagnostic

This section records the post-hoc sequence from parameter inspection to a multi-clamp sweep and broad benchmark re-evaluation. It analyzes released checkpoints rather than a new trained architecture. The primary checkpoint results in Section 5 remain untouched. Every cap below is opt-in, applied only at inference, and removed after each condition.

The diagnostic follows three systematic stages:

1. Parameter-only operating point inspection: Identifying unusually large learned retention gate biases (b_γ) without running the model.
2. Causal multi-clamping sweep: Evaluating multiple clamp values (local percentiles p90, p99 and absolute half-life ceilings hl:256, hl:512) across sequence lengths 2K–256K on clean text loss and needle retrieval.
3. Independent benchmark re-evaluation: Re-benchmarking the selected intervention across short-context downstream tasks, retrieval (synthetic and FinePDFs real-text haystacks, passkey and NIAH), fixed-target long-document likelihood (FinePDFs, PG-19, Proof-Pile), and BABILong reasoning.

I.1 Parameter-only operating points

The memory gate has one learned row per query/memory head, rather than one scalar per attention KV head. For each row we decompose the logit as

$$z_{\gamma,t} = W_\gamma x_t + b_\gamma, \qquad \gamma_0 = \sigma(b_\gamma), \qquad H_{1/2}(\gamma_0) = \frac{\log(1/2)}{\log \gamma_0}. \tag{11}$$

Here $x = 0$ defines a reproducible learned operating point. It is not the empirical mean runtime gate: every reported outlier has a nonzero W_γ row, so activations can shorten or lengthen its instantaneous horizon.

Table 18 reports the parameter operating points across all base checkpoints. The promoted NoPE and Polar checkpoints trained on L40S contain isolated Block 2 heads with learned half-lives of 21.01M and 3.07M tokens. RoPE has a maximum $H_{1/2}$ of 682.2 tokens, while Atma-Raven-Titans ranges from 30.5 to 41.4 tokens. We therefore do not cap Atma-Raven-Titans.

Scanning the separately fine-tuned BABILong checkpoints gives 20.86M tokens for NoPE, 3.05M for Polar, and 680 for RoPE. The outlier therefore remains nearly unchanged after adaptation. Each BABILong cap is still selected from the final fine-tuned checkpoint rather than copied from the base model.

I.2 Paired multi-clamp sweep

For a target maximum half-life $H_{\max}$, the diagnostic clips the final activation-conditioned gate logit,

$$z'_{\gamma,t} = \min\Big(z_{\gamma,t}, \operatorname{logit}(2^{-1/H_{\max}})\Big), \qquad \gamma'_t = \sigma(z'_{\gamma,t}). \tag{12}$$

For hl:256, $H_{\max} = 256 \implies \gamma'_t \leq 0.997296$. Only the single layer/head with the largest parameter-only b_γ is targeted. This upper bound leaves smaller token-conditioned gates unchanged, modifies neither checkpoint tensors nor attention, and is removed after the condition.

Table 19 reports the complete paired sweep across conditions (baseline, p90, p99, hl:256, hl:512) and lengths (2K, 16K, 64K, 256K) within a single process sharing model instance, documents, and seeds:

- Why hl:256 is selected: It gives the largest reduction in 256K needle cross-entropy, from 12.20 to 6.05 nats on NoPE and from 2.39 to 1.71 nats on Polar. It also keeps 2K clean-loss drift within the 0.05-nat guardrail, at +0.022 nats on NoPE and +0.010 on Polar.
- Why p99 fails on saturated checkpoints: On NoPE L40S, the 99th-percentile logit (13.46) still gives a half-life of about 484k tokens. At 256K, clean loss remains at 12.09 nats and needle cross-entropy at 12.15 nats.
- Why RoPE is an empirical negative control: RoPE has no comparable outlier. Capping it at hl:256 gives no retrieval benefit at 256K (0.0% accuracy) and reduces short-context needle accuracy from 82.5% to 50.0%. The intervention is therefore not a general benchmark boost.

I.3 Independent benchmark re-evaluation

After the pilot sweep, we evaluate the capped promoted checkpoints on all eight short-context downstream tasks, the full synthetic and FinePDFs retrieval matrix, dataset-level long-document likelihood, and BABILong QA1–QA10. Base-model and BABILong-fine-tuned checkpoints are pinned to their archived Hugging Face revisions. Direct checkpoint-exact scoring is used throughout. All 15 model–suite jobs completed without an out-of-memory failure.

Tables 18–27 report the full numerical arrays:

- Downstream Controls (Table 20): Unweighted 2K task means change by at most 0.06 points across all three models, verifying short-context preservation.
- Retrieval curves (Tables 21, 22, 23, 24): Polar gains +12.70 points in target-token accuracy and +7.33 points in exact five-token accuracy at 256K. Its 8K FinePDFs exact accuracy rises from 62.7% to 91.3%, and capped exact recall remains nonzero through 128K. NoPE remains at 0.0% exact real-text retrieval from 16K to 256K, and RoPE remains at 0.0% at every extrapolation length.
- BABILong reasoning (Table 25): The cap raises NoPE to 35–57% from 8K to 256K, compared with 0% untouched, and raises Polar from 28% to 42% at 256K. RoPE decreases from 14% to 11%.
- Long-document likelihood (Tables 26, 27): At 256K, the cap reduces NoPE BPB from 11.666 to 1.180 on FinePDFs, from 4.902 to 1.224 on PG-19, and from 7.723 to 2.382 on Proof-Pile. Polar's mean degradation drops from 1.26× to 1.03×.

Checkpoint	Role	Outlier Block/Head	Learned b_γ	Logit z_0	γ_0	Max $H_{1/2}$ (tok)	Median $H_{1/2}$	Min $H_{1/2}$	$\|W_\gamma\|_2$
NoPE (L4, mbs4)	Earlier L4 baseline	Block 2, H7	+1.687	5.587	0.9962667	185.3	29.6	3.9	3.36
NoPE (L40S, mbs4)	Microbatch ablation	Block 2, H2	+10.607	14.507	0.9999995	1.38M	21.8	4.9	3.49
NoPE (L40S, mbs16)	Promoted matched	Block 2, H5	+13.327	17.227	1.0000000	21.01M	26.1	3.5	3.33
Polar (L40S, mbs16)	Promoted matched	Block 2, H6	+11.403	15.303	0.9999998	3.07M	34.3	5.2	2.76
RoPE (L40S, mbs16)	Matched negative control	Block 2, H1	+2.991	6.891	0.9989845	682.2	26.4	8.8	2.19
Atma-Raven-Titans	Recurrent reference	Block 2, H3	+0.182	4.082	0.9834064	41.4	34.9	30.5	1.94

Table 18: Learned zero-input retention operating points across base checkpoints. Isolated Block 2 heads in L40S NoPE and Polar have half-lives of > 3M–21M tokens. The maxima are 682 tokens for RoPE and 41 tokens for Atma-Raven-Titans.

Model	Condition	Clean NLL (nats)				Needle Cross-Entropy (nats)				Needle Acc. (%)	
		2K	16K	64K	256K	2K	16K	64K	256K	2K	256K
NoPE (L40S, mbs16)	Baseline (none)	2.321	2.791	7.177	12.003	0.103	6.105	10.800	12.198	100.0	0.0
	p90 quantile	2.359	1.442	1.263	1.133	0.428	2.511	5.718	6.060	100.0	0.0
	p99 quantile	2.321	2.764	6.955	12.094	0.103	6.016	10.773	12.148	100.0	0.0
	hl:256 (cap) (sel.)	2.343	1.433	1.263	1.138	0.245	2.024	5.687	6.051	100.0	0.0
	hl:512 (cap)	2.333	1.437	1.279	1.162	0.163	1.799	5.651	6.076	100.0	0.0
Polar (L40S, mbs16)	Baseline (none)	2.341	1.423	1.300	1.252	0.262	0.509	1.571	2.387	100.0	50.0
	p90 quantile	2.353	1.399	1.201	1.057	0.282	0.167	0.369	1.713	97.5	65.0
	p99 quantile	2.342	1.422	1.295	1.237	0.264	0.478	1.529	2.366	100.0	55.0
	hl:256 (cap) (sel.)	2.351	1.399	1.201	1.057	0.283	0.165	0.400	1.707	100.0	65.0
	hl:512 (cap)	2.349	1.398	1.201	1.059	0.285	0.163	0.506	1.806	100.0	65.0
RoPE (L40S, mbs16)	Baseline (none)	2.343	2.145	2.581	2.948	0.966	4.809	5.975	6.157	82.5	0.0
	p90 quantile	2.365	2.176	2.488	2.745	3.345	5.053	5.870	5.964	17.5	0.0
	p99 quantile	2.346	2.136	2.478	2.758	1.887	4.919	5.906	6.033	55.0	0.0
	hl:256 (cap)	2.347	2.139	2.477	2.754	2.074	4.936	5.897	6.019	50.0	0.0
	hl:512 (cap)	2.345	2.132	2.485	2.774	1.499	4.886	5.922	6.073	70.0	0.0
NoPE (L40S, mbs4)	Baseline (none)	2.291	5.542	8.425	9.589	0.098	8.093	9.428	10.446	100.0	0.0
	p90 quantile	2.319	1.406	1.250	1.122	0.222	1.283	4.988	5.888	100.0	0.0
	p99 quantile	2.291	5.489	8.168	9.236	0.097	8.066	9.382	10.450	100.0	0.0
	hl:256 (cap)	2.304	1.394	1.274	1.181	0.161	1.167	5.127	6.108	100.0	0.0
	hl:512 (cap)	2.298	1.534	1.664	1.549	0.127	2.248	5.353	6.151	100.0	0.0
NoPE (L4, mbs4)	Baseline (none)	2.304	2.546	3.010	3.615	0.163	1.326	2.649	3.730	97.5	15.0
	p90 quantile	2.307	1.828	2.046	2.333	0.181	1.929	2.694	4.447	100.0	5.0
	p99 quantile	2.306	1.934	2.210	2.613	0.177	1.899	2.699	4.435	97.5	10.0
	hl:256 (cap)	2.305	2.016	2.334	2.776	0.174	1.843	2.706	4.357	97.5	10.0
	hl:512 (cap)	2.304	2.181	2.565	3.030	0.171	1.699	2.719	4.142	97.5	10.0

Table 19: Complete paired clamp sweep across quantile and half-life caps from 2K to 256K. Capping the single outlier head at hl:256 improves NoPE and Polar extrapolation while meeting the 2K short-context guardrail. The p99 setting leaves a long retention horizon, while the RoPE cap reduces short-range needle accuracy from 82.5% to 50.0% without improving long-range retrieval.

Model	LAMBADA	HellaSwag	PIQA	WinoGrande	ARC-E	ARC-C	OBQA	BoolQ	Mean
NoPE	30.16 → 30.22	37.88 → 37.88	66.27 → 66.21	51.93 → 51.38	50.53 → 50.70	32.44 → 32.44	31.60 → 31.60	60.37 → 60.34	45.15 → 45.10
Polar	28.47 → 28.22	36.19 → 36.20	66.87 → 67.30	51.62 → 51.38	49.12 → 48.60	26.09 → 26.42	31.80 → 32.00	56.21 → 55.87	43.29 → 43.25
RoPE	30.25 → 30.16	37.19 → 37.17	66.76 → 66.97	50.67 → 51.07	49.12 → 48.95	29.43 → 29.77	32.40 → 32.20	60.95 → 61.01	44.60 → 44.66

Table 20: Complete short-context control, untouched → capped (accuracy, %). The intervention changes no task mean materially (< 0.06 points delta).

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	99.2 → 98.8	75.8 → 88.7	48.9 → 81.1	47.5 → 46.7	35.2 → 35.5	13.2 → 16.6	5.1 → 9.2	0.6 → 0.9
Polar	98.9 → 98.6	92.4 → 97.8	87.5 → 97.2	73.1 → 77.6	67.8 → 83.5	54.3 → 69.0	41.8 → 63.6	34.4 → 47.1
RoPE	88.2 → 76.1	24.8 → 21.5	17.4 → 13.0	0.4 → 0.0	3.4 → 2.1	0.1 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 21: Complete retrieval curve, untouched → capped, reported as teacher-forced target-token accuracy (%). Each entry averages passkey and NIAH, synthetic and FinePDFs haystacks, and three depths.

These results support an isolated near-unit retention gate as a major mediator of NoPE and Polar extrapolation degradation. They motivate testing bounded retention gates during training, but the inference-only intervention does not validate that design.

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	96.0 → 94.0	55.8 → 75.0	44.3 → 62.2	39.0 → 32.7	11.8 → 9.2	2.3 → 0.5	0.3 → 0.0	0.0 → 0.0
Polar	94.7 → 93.2	79.8 → 89.7	74.5 → 86.5	49.0 → 43.0	38.0 → 44.2	32.8 → 33.0	13.2 → 19.8	9.0 → 16.3
RoPE	59.8 → 45.0	2.2 → 0.8	0.3 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 22: Complete retrieval curve, untouched → capped, reported as exact five-token accuracy (%). Each entry averages passkey and NIAH, synthetic and FinePDFs haystacks, and three depths.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	Synthetic	99.1 → 98.7	98.7 → 97.7	97.7 → 95.8	94.9 → 90.8	70.3 → 69.7	26.5 → 33.2	10.2 → 18.4	1.2 → 1.9
	FinePDFs (Real)	99.3 → 98.8	52.9 → 79.8	0.0 → 66.4	0.0 → 2.7	0.0 → 1.3	0.0 → 0.1	0.0 → 0.0	0.0 → 0.0
Polar	Synthetic	99.1 → 98.5	98.4 → 97.5	97.1 → 96.1	97.0 → 96.0	94.9 → 93.9	92.3 → 92.5	80.3 → 85.5	68.4 → 77.5
	FinePDFs (Real)	98.6 → 98.7	86.3 → 98.1	77.9 → 98.3	49.1 → 59.1	40.7 → 73.1	16.3 → 45.5	3.2 → 41.6	0.3 → 16.7
RoPE	Synthetic	81.7 → 63.4	44.9 → 40.3	34.5 → 25.9	0.8 → 0.1	6.8 → 4.3	0.1 → 0.1	0.0 → 0.0	0.0 → 0.0
	FinePDFs (Real)	94.8 → 88.8	4.7 → 2.7	0.2 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 23: Teacher-forced retrieval token accuracy (%) by haystack type, untouched → capped. Polar gains +9.1% synthetic and +16.3% FinePDFs retrieval at 256K, while NoPE FinePDFs retrieval remains 0.0% despite synthetic signal gains.

Model	Haystack	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	Synthetic	95.3 → 93.7	93.7 → 88.7	88.7 → 79.7	78.0 → 65.3	23.7 → 18.3	4.7 → 1.0	0.7 → 0.0	0.0 → 0.0
	FinePDFs (Real)	96.7 → 94.3	18.0 → 61.3	0.0 → 44.7	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0
Polar	Synthetic	96.3 → 92.7	92.3 → 88.3	86.3 → 81.7	86.0 → 81.0	76.0 → 72.0	65.7 → 65.3	26.3 → 38.7	18.0 → 32.7
	FinePDFs (Real)	93.0 → 93.7	67.3 → 91.0	62.7 → 91.3	12.0 → 5.0	0.0 → 16.3	0.0 → 0.7	0.0 → 1.0	0.0 → 0.0
RoPE	Synthetic	42.3 → 27.0	4.3 → 1.7	0.7 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0
	FinePDFs (Real)	77.3 → 63.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0	0.0 → 0.0

Table 24: Exact five-token retrieval accuracy (%) by haystack type, untouched → capped. On real text, RoPE reaches 0.0% at 4K and NoPE at 8K. Polar scores 62.7% untouched and more than 91% capped at 8K, with nonzero capped accuracy through 128K.

Model	0K	1K	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	63 → 63	57 → 60	55 → 55	56 → 56	39 → 57	21 → 54	6 → 45	0 → 37	0 → 35	0 → 39
Polar	55 → 55	54 → 54	52 → 52	54 → 54	56 → 57	50 → 50	46 → 43	45 → 44	35 → 44	28 → 42
RoPE	59 → 61	67 → 68	64 → 66	43 → 50	28 → 33	19 → 21	15 → 17	16 → 15	14 → 13	14 → 11

Table 25: Complete BABILong curve after adaptation, untouched → capped (macro exact match, %). The cap raises NoPE to 35–57% across 8K–256K, compared with 0% untouched, and raises Polar at 256K from 28% to 42%.

Model	2K	4K	8K	16K	32K	64K	128K	256K
NoPE	1.432 → 1.454	1.717 → 1.454	2.656 → 1.473	3.374 → 1.484	5.527 → 1.504	5.885 → 1.506	6.369 → 1.530	8.097 → 1.595
Polar	1.451 → 1.458	1.442 → 1.451	1.406 → 1.407	1.417 → 1.384	1.500 → 1.399	1.569 → 1.400	1.690 → 1.449	1.825 → 1.501
RoPE	1.439 → 1.440	1.642 → 1.656	1.783 → 1.783	1.935 → 1.932	2.094 → 2.082	2.252 → 2.220	2.413 → 2.378	2.512 → 2.460

Table 26: Complete fixed-target likelihood curve, untouched → capped, averaged over FinePDFs, PG-19, and Proof-Pile in mean bits per byte (lower is better).

The complete machine-readable artifacts are under gamma_diagnostics/results/: parameters/gamma_parameters.json for the scan, all_clamp_sweep_262k.json for the paired pilot, and re_evaluation/run-summary.json plus per-suite logs for the full benchmark. The executable protocol and reporting caveats are in gamma_diagnostics/README.md.

J Stress and Environment Diagnostics

Figure 7 reports the sampled gains. For each block f_i, the diagnostic samples

$$G_i = \frac{\|f_i(x + \delta x) - f_i(x)\|_{\mathrm{rms}}}{\|\delta x\|_{\mathrm{rms}}}, \qquad \|\delta x\|_{\mathrm{rms}} = 0.02\|x\|_{\mathrm{rms}}. \tag{13}$$

Model	Dataset	2K	4K	8K	16K	32K	64K	128K	256K
	FinePDFs	0.913 → 0.924	0.868 → 0.885	1.366 → 0.900	1.473 → 0.920	5.317 → 0.969	5.519 → 1.010	6.637 → 1.060	11.666 → 1.180
NoPE	PG-19	1.197 → 1.211	1.461 → 1.216	2.191 → 1.207	2.473 → 1.200	4.259 → 1.205	4.799 → 1.207	4.927 → 1.214	4.902 → 1.224
	Proof-Pile	2.186 → 2.226	2.821 → 2.262	4.411 → 2.312	6.177 → 2.331	7.006 → 2.338	7.337 → 2.300	7.544 → 2.315	7.723 → 2.382
	FinePDFs	0.946 → 0.950	0.913 → 0.913	0.929 → 0.925	0.938 → 0.901	1.032 → 0.924	1.113 → 0.967	1.252 → 1.011	1.419 → 1.052
Polar	PG-19	1.226 → 1.229	1.224 → 1.227	1.221 → 1.215	1.229 → 1.212	1.259 → 1.222	1.287 → 1.243	1.331 → 1.277	1.374 → 1.319
	Proof-Pile	2.181 → 2.196	2.190 → 2.212	2.067 → 2.079	2.084 → 2.040	2.208 → 2.050	2.308 → 1.989	2.487 → 2.059	2.683 → 2.130
	FinePDFs	1.036 → 1.040	1.277 → 1.303	1.492 → 1.486	1.802 → 1.772	2.142 → 2.071	2.469 → 2.353	2.757 → 2.626	2.919 → 2.742
RoPE	PG-19	1.204 → 1.202	1.317 → 1.316	1.350 → 1.351	1.419 → 1.426	1.472 → 1.476	1.541 → 1.530	1.591 → 1.586	1.626 → 1.627
	Proof-Pile	2.077 → 2.078	2.331 → 2.350	2.507 → 2.513	2.582 → 2.596	2.668 → 2.700	2.746 → 2.776	2.892 → 2.922	2.991 → 3.011

Table 27: Dataset-level fixed-target long-document likelihood, untouched → capped, in bits per byte (lower is better). At 256K, the cap reduces NoPE BPB from 11.666 to 1.180 on FinePDFs, from 4.902 to 1.224 on PG-19, and from 7.723 to 2.382 on Proof-Pile.

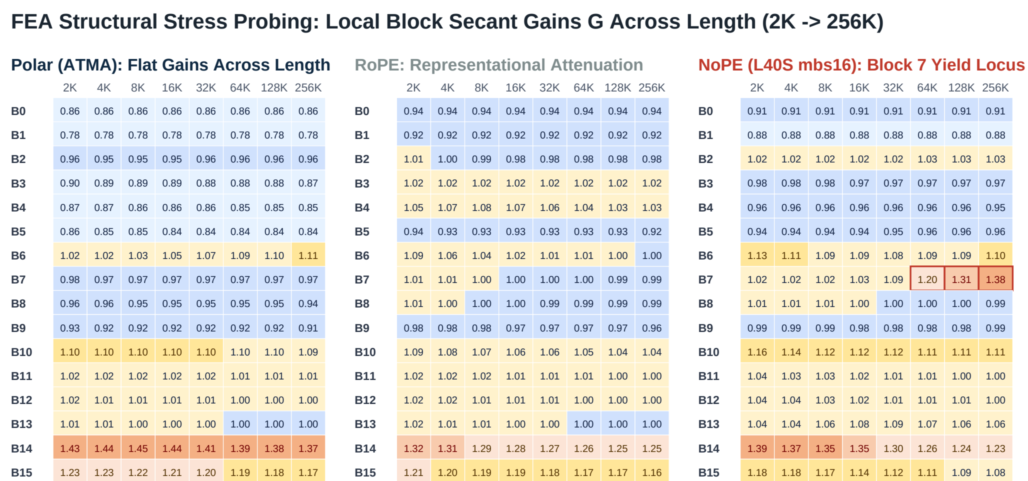

Figure 7: Block-wise sampled secant gains for the three matched attention checkpoints over 64 nested FinePDFs documents and 64 perturbation directions per document. Raven Native and Atma-Raven-Titans were not run under this checkpoint-specific probe. The diagnostic localizes changes but does not estimate a global Lipschitz constant.

The maximum sampled gain is a lower bound on the local operator norm along sampled directions. It is not a tight bound on the block Lipschitz constant. In the observed NoPE collapse, changes first concentrate near blocks 6–7 and later propagate through the residual stream. Polar's sampled gains and activation moments remain comparatively flat through 256K.

K Checkpoint-Variability Diagnostic Journey

The variability finding emerged sequentially. Stage I used L4 GPUs, followed by the first 10B NoPE control on an L4. The difference became visible after primary training moved to L40S. Short-context convergence remained nearly identical (< 0.01% loss difference), while 256K NLL differed (4.07 vs 10.77 nats). Table 28 records the seven-step diagnostic sequence and the retention-horizon intervention.

This sequence localizes the observed checkpoint failure to one unusually long zero-input retention horizon, and the inference cap reverses much of the measured degradation. It does not establish what training event produced the outlier or whether hardware-level differences contributed to it. All five headline Stage II models were trained in the same L40S environment, providing an internally controlled comparison for those runs.

Step	Observation or intervention	Diagnostic implication
1. L4 baseline	Stage I and first 10B NoPE control on L4 (mbs4); 256K NLL was 4.07.	Established baseline extrapolation, but lacked replicated device-class controls.
2. L40S transition	Promoted NoPE run on L40S (mbs16); 2K validation matched (2.8126 vs 2.8160), but 256K NLL rose to 10.77.	Showed that the short-context pretraining curves did not predict long-context behavior.
3. Evaluation replay	L4 and L40S checkpoints re-evaluated under PyTorch 2.12 and 2.13 with identical metrics.	Excluded evaluation software runtime mismatch as the source.
4. Microbatch control	NoPE retrained on L40S with mbs4; 2K validation ended at 2.8128 and 256K NLL at 9.90.	Excluded microbatch size alone as the explanatory mechanism.
5. Deterministic fixtures	Synthetic 2K stream fixtures across device classes showed zero gradient or loss divergence.	Confirmed short-context kernel execution consistency; left training trajectory unconstrained.
6. Retention parameter inspection	Evaluated learned zero-input operating point $\gamma_0 = \sigma(b_\gamma)$ across all layers and heads.	Identified an outlier: L40S NoPE developed an isolated Block 2 head with $b_\gamma = +13.33$ ($H_{1/2} = 21.01$M tokens), whereas L4 NoPE had max $H_{1/2} = 185.3$ tokens.
7. Single-head causal intervention	Installed a reversible inference cap ($H_{\max} = 256$ tok) on the single outlier head and re-evaluated 2K–256K.	Tested the mechanism: Capping the outlier head reduced 256K NLL from 12.00 to 1.14 nats and raised 256K BABILong to 39%, supporting excessive retention as a mediator of the collapse.

Table 28: Diagnostic sequence from the initial L4/L40S divergence to the retention-horizon intervention. Steps 1–5 ruled out evaluation artifacts and microbatch size. Steps 6–7 identified the isolated retention-gate outlier and tested its effect.